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The balance between microbial arsenic methylation and demethylation in paddy soils underpins global arsenic risk and straighthead disease in rice

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Arsenic contamination in rice poses a global challenge to food safety and agricultural productivity, as toxic methylated arsenic species—dimethylarsinic acid (DMA) and its highly toxic derivative, methylated monothioarsenate (DMMTA)—accumulate in rice grains. These arsenic species endanger human health and trigger rice straighthead disease, a crop disorder that drastically reduces yields. However, the microbial ecological processes driving arsenic speciation in paddy soils, and their link to striking geographic disparities in rice arsenic speciation profiles and disease prevalence, remain poorly understood. Here, we integrate soil chronosequences spanning 1 to 2,000 y of rice cultivation, a global metagenomic survey of 801 paddy soils, controlled incubations, and field surveys to demonstrate that the balance between arsenic-methylating and arsenic-demethylating microbes is the key determinant of rice grain arsenic speciation and straighthead disease susceptibility. We show that young and moderate-age paddy soils (<700 y), common in regions such as the Americas and Europe, are enriched in arsenic-methylating bacteria, leading to elevated DMA and DMMTA in soils and rice grains. In contrast, ancient paddies in Southeast Asia harbor robust populations of DMA-demethylating methanogenic archaea that effectively mitigate the buildup of these toxic arsenic species. We identify core microbial taxa whose abundances serve as predictive biomarkers and construct a global risk map linking a high methylator-to-demethylator ratio in soils with increased straighthead disease incidence. These findings advance our understanding of arsenic biogeochemistry in agroecosystems and establish a predictive framework for identifying regions at elevated risk of arsenic-induced crop disorders and food contamination.

arsenic | paddy soil | rice straighthead disease | microbial ecology | food safety

Significance

Rice straighthead disease, triggered by toxic methylated arsenic species, poses a major yet underappreciated threat to global rice yields and food safety. Here, we reveal that the balance between microbial arsenic methylators and demethylators in paddy soils—shaped by cultivation history—determines grain arsenic speciation and rice disease risk. We identify key microbial taxa for arsenic methylation and demethylation and map global hotspots of rice straighthead disease. Our study provides microbial indicators to predict rice arsenic speciation and straighthead disease risk with broad implications for global food safety and sustainable rice production.

Arsenic contamination in paddy soils poses an escalating threat to global food safety and sustainable rice production, given that rice is a staple for over half the world's population (1–4). In rice grains, arsenic is present as inorganic arsenic (iAs) and methylated arsenic species (mAs) (5), including dimethylarsinic acid (DMA) and its highly toxic derivative, methylated monothioarsenate (DMMTA) (6, 7). While iAs have received regulatory attention due to its carcinogenicity, DMMTA—which has been recently detected in rice globally (6)—is 3 to 10 times more cytotoxic in human and animal cells (8, 9) and is also highly phytotoxic. In contrast, DMA, although less harmful to humans (10), is a well-known plant toxin which has historically been used as a herbicide and defoliant (11), and is now recognized as a causal factor of rice straighthead disease (12, 13). Straighthead is a physiological disorder characterized by distorted "parrot beak" husks and can result in up to 70% yield losses in severely affected regions (14–17). Recent findings have also found that DMMTA is a potent inducer of straighthead disease, compounding the food security threat (18). Alarmingly, climate change is projected to exacerbate arsenic accumulation in rice grains and associated yield losses (3).

Globally, the proportion of arsenic present as DMA and DMMTA in rice grains varies widely, ranging from ~10 to 90% (DMMTA 1 to 21%) of total grain arsenic (19, 20). Regions with shorter rice cultivation histories, for example in North America and Europe, consistently exhibit higher fractions of these toxic methylated species in rice, compared to the ancient paddies of Southeast Asia (6, 19). Similar geographic disparities are observed within China (21, 22). Areas with elevated methylated arsenic in rice tend to suffer more frequent straighthead disease outbreaks. Intriguingly, straighthead disease often manifests in newly established rice fields and in paddies rotated with upland crops (12, 23), even when total soil arsenic remains within natural background levels (24). These patterns highlight a critical and often overlooked point: Total arsenic concentration alone is a poor predictor of food safety or disease risk. Instead, arsenic speciation is the dominant factor

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determining both grain arsenic species and straighthead disease, emphasizing the need for speciation-focused risk assessments in rice agroecosystems.

Rice plants themselves lack the capacity for arsenic methylation; the DMA and DMMTA present in rice plants are predominantly derived from soil microbial process (25). Soil microorganisms, particularly those harboring arsenite *S*-adenosylmethionine methyltransferase (ArsM) (26), are key producers of DMA (27–29). The dynamic redox conditions of flooded paddy soils drive an interplay between arsenic methylation and demethylation (23), mediated primarily by anaerobic microbes: sulfate-reducing bacteria and other anaerobes catalyze arsenic methylation (24, 28, 30, 31), while methanogenic archaea identified by the presence of methylcoenzyme M reductase (McrA) facilitate DMA demethylation (23, 32). However, which specific microbial taxa are the dominant "arsenic methylators" and "arsenic demethylators," and how their activities explain the pronounced geographic differences in rice arsenic speciation and straighthead disease prevalence, remain unresolved. This knowledge gap hinders the development of effective strategies to predict and manage arsenic-related risks in rice production.

We hypothesize here that the global patterns of rice arsenic speciation and straighthead disease are governed by the balance between arsenic-methylating and -demethylating microbes, a balance that is further modulated by paddy soil cultivation history and agronomic practices. For instance, relatively young paddy soils in the Americas, Europe, and Northeast China, established within the past 100 to 400 y (33, 34), may favor DMA production due to a higher prevalence of methylators, whereas ancient paddies in Southeast Asia may be enriched in methanogenic archaea (35) that enhance DMA demethylation. Similarly, converting upland fields to paddy or practicing paddy-upland crop rotations could transiently disrupt the microbial equilibrium and promote straighthead disease outbreaks (36).

To test these hypotheses, we integrated soil chronosequence experiments spanning 1 to 2,000 y, controlled incubations, extensive field surveys, and a global metagenomic analysis of 801 paddy soil samples. Our results reveal that soil age critically shapes microbial communities: Young and moderate-age paddy soils (i.e., younger than 700 y) are enriched in arsenic methylators, leading to elevated levels of DMA and DMMTA and higher disease susceptibility, whereas older soils (i.e., older than 700 y) sustain robust populations of DMA-demethylating methanogens that mitigate these toxic species. By linking microbial community dynamics with soil age and agronomic practices, our work provides a mechanistic framework to explain the geographic disparities in rice arsenic speciation and the incidence of straighthead disease, with profound implications for global food safety and sustainable agriculture.

Results

Paddy Soil Age Shapes Arsenic Speciation in Rice Grains. We first surveyed three representative paddy soil chronosequences (long-term rice cultivation sequences) in China with cultivation ages from ~1 y up to ~2,000 y (Fig. 1 *A–D*). Across these chronosequences, total arsenic concentrations in soil remained relatively constant with age [e.g., 9.3 to 12.5 mg kg^{-1} in Yingtan (YT), 11.6 to 13.5 mg kg^{-1} in Guang'an (GA), and 5.7 to 9.7 mg kg^{-1} in Cixi (CX)]. In contrast, soil organic carbon content increased steadily with cultivation age (*SI Appendix*, Table S1), consistent with progressive carbon accumulation in long-term paddies (37).

Analysis of arsenic species in rice grains revealed that arsenite—present primarily as the neutral species H_3AsO_3 under pH <9.2—was the dominant form of inorganic arsenic (iAs) in both rice husks and brown rice, with concentrations of 216 to 309 μg kg^{-1} in husks and 78 to 109 μg kg^{-1} in brown rice. Methylated As (mAs) species, predominantly DMA and DMMTA, were present at concentrations of 34 to 167 μg kg^{-1} in husks and 6.5 to 42 μg kg^{-1} in brown rice (Fig. 1*E* and *SI Appendix*, Fig. S1). Although these concentrations are lower than iAs, the presence of DMMTA (even at μg kg^{-1} levels) is notable given its high toxicity.

Crucially, the proportion of arsenic present as methylated species increased with cultivation age in the two younger chronosequences. In YT (max 100 y cultivation), mAs in husks rose from 14% at 1 y to 38% at 100 y (and from 6.8 to 34% in brown rice). In GA (max 300 y), mAs in husks went from 24% at 1 y to 32% by 300 y (20 to 29% in brown rice). However, in the oldest chronosequence (CX, up to ~2,000 y), mAs proportions peaked at intermediate ages (27% in husks and 26% in brown rice at 500 y) and then declined to 21% (husks) and 16% (brown rice) by 2,000 y (Fig. 1*E* and *SI Appendix*, Fig. S1). Thus, the younger paddies tended to accumulate more DMA/DMMTA in rice, whereas the very old paddies showed a relative decline in those species. Together, these field results suggest that the duration of rice cultivation significantly influences arsenic speciation in rice grains—younger paddy soils promote methylated arsenic accumulation in rice while sufficiently old paddies appear to curb it.

Cultivation Age Affects the Dynamics of Methylated Arsenic in Flooded Soil. Because mAs in rice originate from paddy soils (25, 38), we next examined how paddy soil age influences arsenic dynamics in the soil porewater under flooded conditions. We conducted anaerobic microcosm incubations using soils from each chronosequence (Fig. 2). Upon flooding and incubation, all soils rapidly became anaerobic: The redox potential (Eh) dropped sharply to around −150 mV, though younger soils took slightly longer to reach strongly reducing conditions (*SI Appendix*, Fig. S2 *A–C*). By the end of the 28-day incubation, porewater pH in all samples stabilized near neutral (*SI Appendix*, Fig. S2 *D–F*). As expected under anoxic conditions, iron and sulfate reduction occurred; dissolved Fe in porewater increased while sulfate concentrations declined (*SI Appendix*, Fig. S2 *G–O*), with these redox changes happening faster in older soils than in younger soils. We also observed increasing production of dissolved organic carbon (DOC) and methane (CH_4) over time, especially in soils with longer cultivation histories (*SI Appendix*, Fig. S2 *P–U*), indicating more active microbial decomposition in older paddy soils.

Under these flooded conditions, arsenic was mobilized into porewater. Total dissolved arsenic (tAs) and dissolved iAs concentrations rose during incubation, reaching peaks that varied with soil age. In younger chronosequence soils (YT, GA), the amount of arsenic released into porewater increased consistently with cultivation age. In the oldest chronosequence (CX), arsenic release also increased up to about 700 y and then declined at the 2,000-y soil (Fig. 2 *A–F*). This pattern mirrors the trends in dissolved Fe (*SI Appendix*, Fig. S2 *G–I*), suggesting that reductive dissolution of iron oxides (which can adsorb arsenic) is a primary driver of arsenic release into porewater across different soil ages.

Throughout the incubations, the dominant arsenic species in porewater were iAs and DMA, accounting for approximately 51 to 89% and 2.3 to 34% of total dissolved As, respectively (Fig. 2 *E–J*). Notably, porewater DMA showed a transient behavior; DMA concentrations increased during the first 7 to 14 d of flooding and then disappeared by the end of 28 d in all soils (Fig. 2 *H–J*). In soils younger than ~700 y, DMA peaked at day 7, with higher peak values in soils of greater cultivation age. In the >700-year-old soil (oldest CX), the DMA peak was delayed to day 14 and was lower in magnitude (Fig. 2*J*). The toxic methylated thioarsenic species (i.e., DMMTA and DMDTA, both derived

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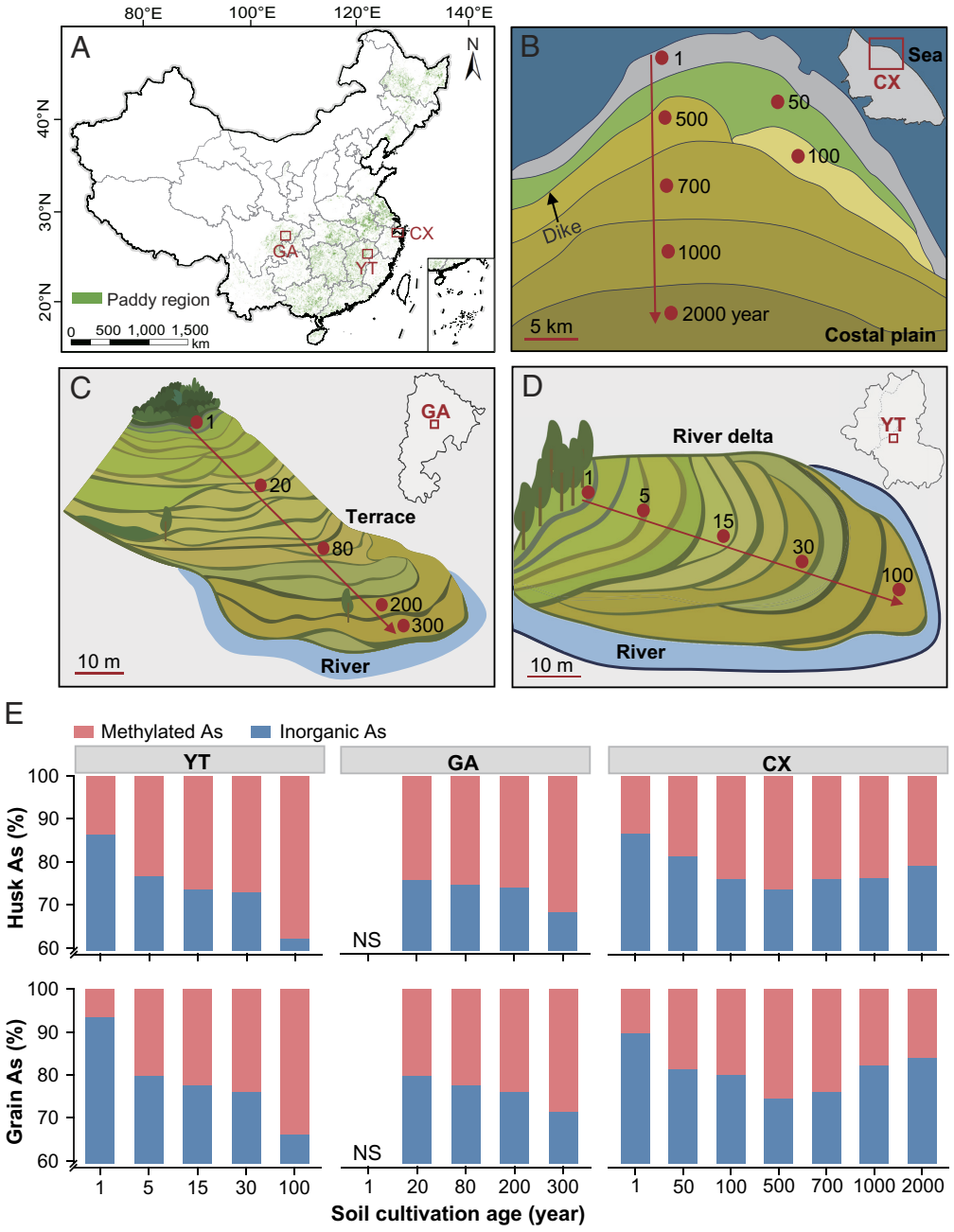


Fig. 1. Geographic locations of three paddy soil chronosequences with varying cultivation ages and arsenic species distribution in rice husk and brown rice. (*A*) Map indicating the locations of three paddy soil chronosequences. (*B*) Cixi (CX): paddy soils reclaimed from the sea, with cultivation ages ranging from 1 to 2,000 y. (*C*) Guang'an (GA): terraced paddy soils, with cultivation ages increasing from the summit to the foot of the hill, spanning 1 to 300 y. (*D*) Yingtan (YT): paddy soils reclaimed from forest near a river, with cultivation ages decreasing from the river to the forest, ranging from 1 to 100 y. (*E*) Proportions of inorganic As [sum of arsenite (As(III)) and arsenate (As(V))] and methylated As [sum of dimethylarsinic acid (DMA) and demethylated monothioarsenate (DMMTA)] in rice husk and brown grain. NS indicates that no rice was planted in the field.

from DMA) followed similar temporal patterns as DMA (*SI Appendix*, Fig. S3). These incubation results indicate that paddy soil age strongly influences the production and fate of methylated arsenic in anoxic soils.

Arsenic Methylation and Demethylation Rates Vary With Soil Age. Previous studies have shown that the decline in porewater DMA concentration during the latter phase of incubation is caused by microbial demethylation (24). Using the incubation data and kinetic modeling (*SI Appendix*, Fig. S4 and Tables S2–S5), we calculated the rates of arsenic methylation and DMA demethylation (Fig. 3 *A* and *B*). We found that all data from different soils and cultivation ages can be fitted into a single kinetic model for a given incubation time point. At the 7th day, As methylation was the main driver of DMA accumulation across all soils (Fig. 3*A*), as evidenced by increasing porewater DMA concentrations over time (Fig. 2 *H–J*). The As methylation rate exhibited a rapid increase during the first 700 y of paddy soil cultivation, followed by little further increase between 700 and 2,000 y (Fig. 3*A*). In contrast, the DMA demethylation rate steadily increased from 1 to 2,000 y of paddy soil cultivation. In soils younger than 700 y, the net DMA accumulation rate (methylation rate minus demethylation rate) increased with the paddy soil cultivation age, leading to an overall net increase in DMA in porewater (Fig. 3*A* and *SI Appendix*, Fig. S5*B*). However, in soils older than 700 y, the net accumulation rate declined with the cultivation age, resulting in a gradual reduction in porewater DMA concentrations (Fig. 2*J*).

By the 14th day, DMA demethylation became the dominant factor controlling DMA accumulation across all soils (Fig. 3*B*), as indicated by decreasing porewater DMA concentration (Fig. 2 *H–J*). Although both the As methylation and DMA demethylation rates increased with the soil cultivation age, demethylation surpassed methylation, resulting in increased DMA consumption (i.e., methylation < demethylation) and lower porewater DMA concentrations (Fig. 3*B* and *SI Appendix*, Fig. S5*D*).

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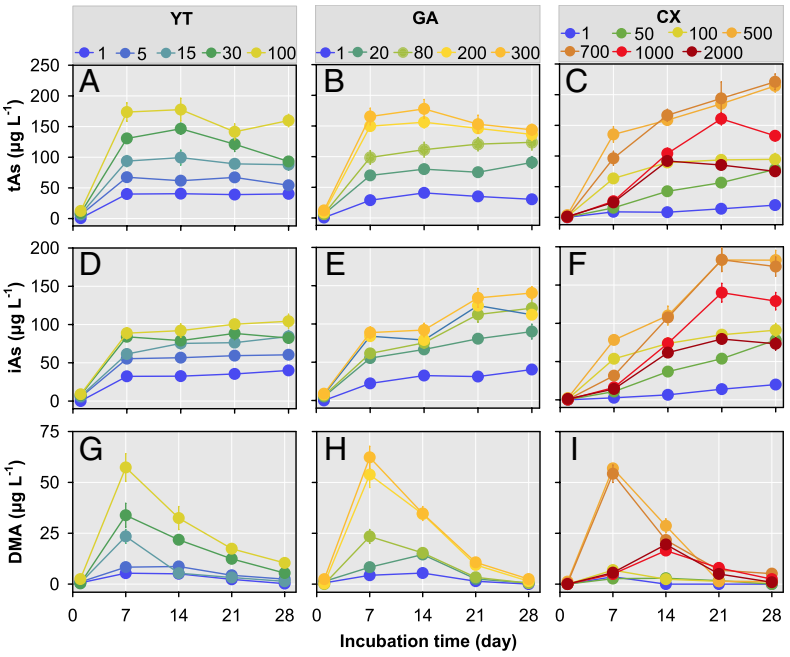


Fig. 2. Soil microcosm incubation of three paddy soil chronosequences and dynamics of arsenic species in soil porewaters under flooded conditions. (*A–C*) Temporal changes in total dissolved As (tAs), (*D–F*) inorganic As [sum of arsenite (As(III)) and arsenate (As(V))], and (*G–I*) DMA in soil porewaters from the YT (*Left* column), GA (*Middle* column), and CX (*Right* column) chronosequences over a 4-wk flooding period. Data are presented as means ± SE (n = 3).

Arsenic methylation in soil is carried out primarily by microorganisms possessing *arsM* genes (26), whereas some methylotrophic methanogens, carrying *mcrA* genes, contribute to DMA demethylation (24, 32). Therefore, we quantified the abundances of *arsM* and *mcrA*, as well as total bacterial and archaeal 16S rRNA genes and other genes encoding dissimilatory sulfite reductase (*dsr*), respiratory arsenate reductase (*arrA*), and detoxification arsenate reductase (*arsC*), via quantitative PCR. Gene abundances generally increased over incubation time and were positively correlated with soil cultivation age (*SI Appendix*, Fig. S6).

Correlation analyses (Pearson's r) at day 7 revealed that the arsenic methylation rate was strongly and positively correlated with factors such as soil cultivation age, porewater iAs concentration, and soil *arsM* gene abundance (Fig. 3*C*). Similarly, the DMA demethylation rate at day 7 correlated significantly with cultivation age, headspace CH_4 (a proxy for methanogenic activity), and the abundances of *arsM* and *mcrA* genes, as well as porewater iAs and DMA concentrations. By day 14, many correlations with methylation rate had weakened (since net methylation slowed by then), but correlations involving demethylation had strengthened—notably between demethylation rate and cultivation age, CH_4, and *mcrA* gene abundance (Fig. 3*D*). This pattern indicates that over time, the influence of demethylating microbes (methanogens) becomes more pronounced, aligning with the observed late-stage DMA removal.

We employed structural equation modeling (SEM) to elucidate the causal pathways by which soil cultivation age influences arsenic methylation dynamics. The analysis revealed that the dominant pathway begins with soil cultivation age driving accumulation of soil organic carbon (SOC), which in turn regulates the abundances and composition of functional genes (*arsM* and *mcrA*), ultimately controlling the rates of arsenic methylation and demethylation and the net accumulation of methylated arsenic species (Supplementary Fig. S7). The SEM models accounted for 97% and 92% of the variance in net DMA accumulation rates at days 7 and day 14, respectively. Specifically, on day 7, cultivation age significantly affected porewater DOC concentrations, which positively influenced *arsM* gene abundance, thereby promoting arsenic methylation and DMA accumulation (Fig. 3*E*). By day 14, DOC continued to mediate microbial responses but primarily impacted *mcrA* gene abundance, facilitating enhanced DMA demethylation and reducing net DMA accumulation (Fig. 3*F*). These findings demonstrate that long-term cultivation shapes soil biogeochemistry, particularly organic carbon availability, which in turn structures microbial functional gene pools and governs arsenic transformation pathways in paddy soils.

Core Arsenic-Methylating Microbes in Paddy Soils. Having established the connection between soil age, microbial genes, and arsenic transformation rates, we next sought to identify the key microbial players—the core taxa responsible for arsenic methylation and for DMA demethylation in these soils. We performed shotgun metagenomic sequencing on the incubation soil samples (from all chronosequences at days 7 and 14). This yielded a total of 8102 distinct *arsM* gene sequences belonging to microbes across 31 different phyla. The majority of these *arsM* genes were found in bacteria, notably members of *Acidobacteria*, *Actinobacteria Armatimonadota*, *Bacillota, Bacteroidetes*, *Chloroflexota*, *Firmicutes*, *Gemmatimonadetes*, *Proteobacteria*, *Pseudomonadota*, and *Verrucomicrobiota* (*SI Appendix*, Fig. S8). This indicates that a diverse array of bacteria in paddy soils carry the genetic potential to methylate arsenic.

To pinpoint which of these were most important, we correlated each *arsM* gene's abundance with the observed arsenic methylation rates. We found 213 *arsM* genes whose abundance was significantly positively correlated with arsenic methylation rate at day 7. Moreover, the abundances of these 213 genes increased consistently with soil cultivation age (Fig. 4 *A–C*), suggesting they thrive in older-cultivated soils. These genes were hosted by bacteria from the phyla listed above. Within each chronosequence, dozens of *arsM* genes correlated with methylation (99 genes in YT, 81 in GA, 93 in CX; Fig. 4*D*). Importantly, 11 specific *arsM* genes were shared among all three chronosequences, which we designate as the putative "core arsenic methylators" (Fig. 4 *D* and *E*). The combined relative abundance of these 11 core *arsM* genes ranged from ~0.02% up to 4.7% of the metagenomes, and they belong to diverse phyla (*Firmicutes*, *Proteobacteria*, *Acidobacteriota*, *Gemmatimonadetes*, *Verrucomicrobiota*). Interestingly, total *arsM* gene diversity (richness) did not change with soil age (*SI Appendix*, Fig. S9 *A–C*) and was not directly correlated with methylation rates (*SI Appendix*, Fig. S9*D*), implying that it is the presence of specific key *arsM* genes,

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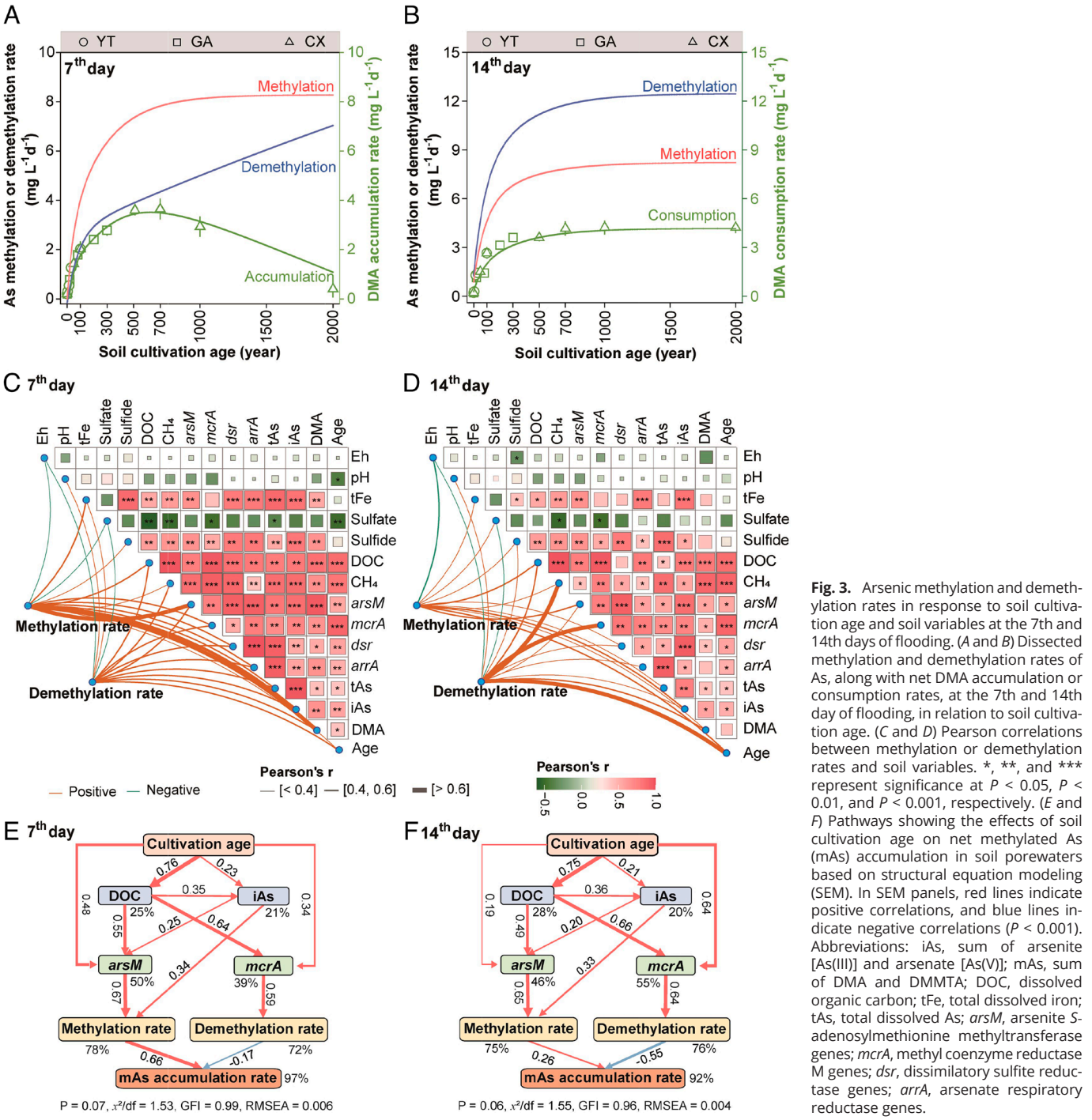


Fig. 3. Arsenic methylation and demethylation rates in response to soil cultivation age and soil variables at the 7th and 14th days of flooding. (*A* and *B*) Dissected methylation and demethylation rates of As, along with net DMA accumulation or consumption rates, at the 7th and 14th day of flooding, in relation to soil cultivation age. (*C* and *D*) Pearson correlations between methylation or demethylation rates and soil variables. *, **, and *** represent significance at $P < 0.05$, $P < 0.01$, and $P < 0.001$, respectively. (*E* and *F*) Pathways showing the effects of soil cultivation age on net methylated As (mAs) accumulation in soil porewaters based on structural equation modeling (SEM). In SEM panels, red lines indicate positive correlations, and blue lines indicate negative correlations ($P < 0.001$). Abbreviations: iAs, sum of arsenite [As(III)] and arsenate [As(V)]; mAs, sum of DMA and DMMTA; DOC, dissolved organic carbon; tFe, total dissolved iron; tAs, total dissolved As; *arsM*, arsenite *S*-adenosylmethionine methyltransferase genes; *mcrA*, methyl coenzyme reductase M genes; *dsr*, dissimilatory sulfite reductase genes; *arrA*, arsenate respiratory reductase genes.

and not simply having more *arsM* diversity, that drives higher methylation.

Consistently, the aggregate relative abundance of those 213 "methylator" genes was strongly correlated with the actual arsenic methylation rate ($r = 0.77$, $P < 0.001$). More striking, the combined abundance of the 11 core *arsM* genes had an even stronger correlation ($r = 0.84$, $P < 0.001$) with the methylation rate (Fig. 4 *F* and *G*). This correlation remained robust (though slightly lower) at day 14 as well ($r = 0.72$, $P < 0.001$; *SI Appendix*, Fig. S10). Thus, the 11 core arsenic-methylating microbes we identified can serve as reliable indicators (or biomarkers) of a soil's arsenic methylation potential.

Core Arsenic-Demethylating Microbes in Paddy Soils. We carried out a similar analysis for DMA demethylating microbes using the *mcrA* gene. In total, we detected 249 distinct *mcrA* gene sequences in the soils (after 14 d incubation; *SI Appendix*, Fig. S11), affiliated with six genera of methanogenic archaea: *Methanobacterium, Methanocella, Methanofollis, Methanosaeta, Methanosarcina,* and *Methanomassiliicoccus*. Out of these, the abundance of 124 *mcrA* genes showed significant positive correlations with measured DMA demethylation rates (Fig. 5 *A*–*C*). Breaking it down, 35 *mcrA* genes correlated with demethylation in YT, 49 in GA, and 58 in CX soils (Fig. 5*D*). Six *mcrA* genes were common to all three chronosequences, representing the "core DMA demethylators". The combined relative abundance of these six core demethylator genes ranged from ~0.001 to 4.9% of the *mcrA* gene pool (Fig. 5*E*). Both the total pool of demethylator genes (all 124) and the six core genes individually showed strong correlations with demethylation rates, but as with the methylators, the core six gave the strongest indicator ($r = 0.93$, $P < 0.001$ for core six vs $r = 0.85$, $P < 0.001$

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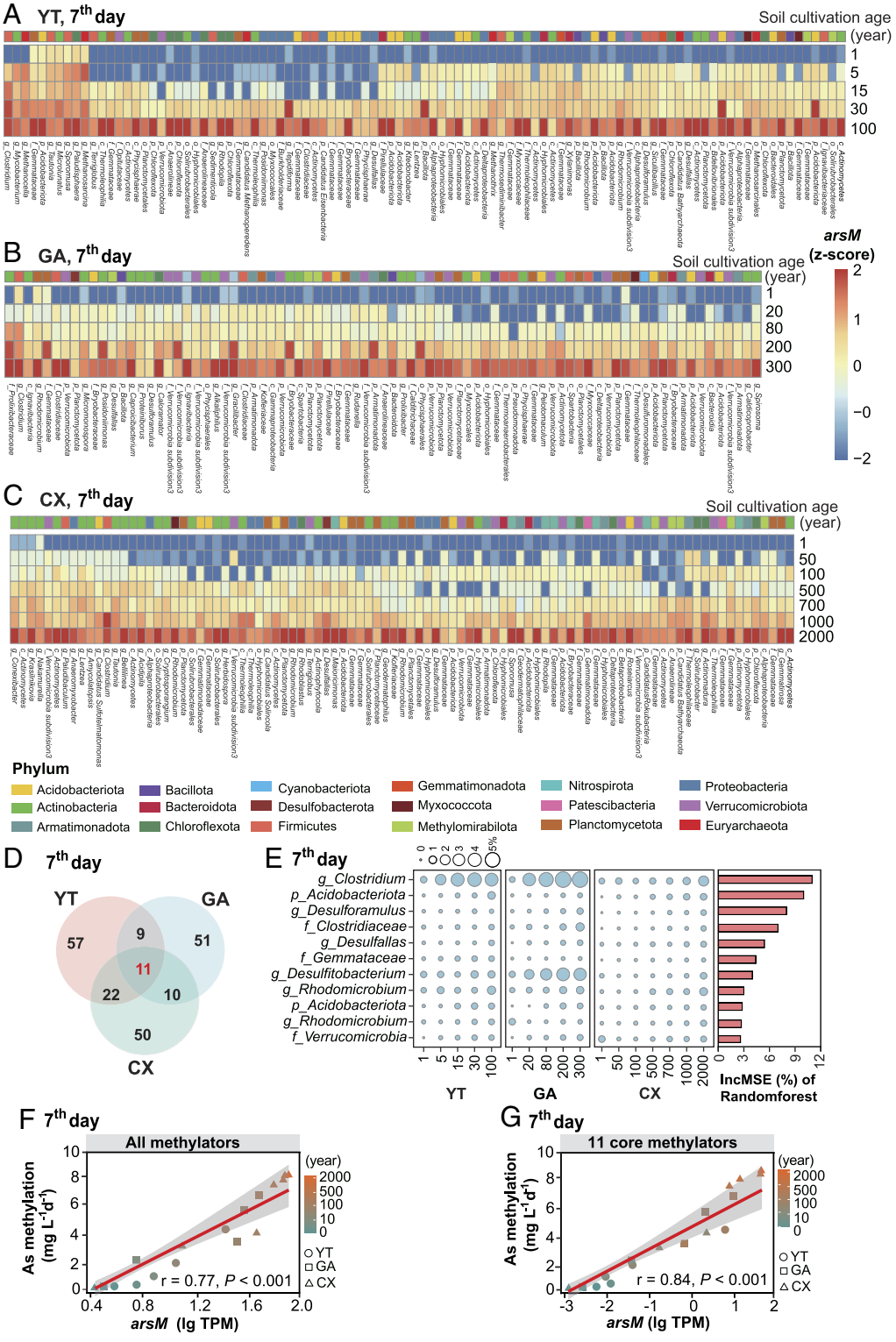


Fig. 4. Core microbial arsenic methylators in three paddy soil chronosequences and their correlations with arsenic methylation rates at the 7th day of flooding. (*A*–*C*) Identified As methylators containing arsenite *S*-adenosylmethionine methyltransferase genes (*arsM*) with abundance positively correlated ($P < 0.05$) with As methylation rates in the YT (*A*), GA (*B*), and CX (*C*) paddy soil chronosequences at the 7th day of flooding. (*D*) Venn diagram showing the number of identified As methylators and the core microbial methylators common to all three paddy chronosequences. (*E*) Relative abundances of these 11 core As methylators and their importance to As methylation rates, as determined by a random forest model. (*F* and *G*) Correlations between the total identified As methylators (*F*) and the 11 core As methylators (*G*) with As methylation rates at the 7th day of flooding.

for all 124; Fig. 5 *F* and *G*). Unlike *arsM*, the diversity of *mcrA* genes increased with soil age (older soils had a broader variety of methanogens; *SI Appendix*, Fig. S12 *A*–*C*), and *mcrA* diversity was positively correlated with demethylation rates ($r = 0.77$ to 0.82, $P < 0.001$; *SI Appendix*, Fig. S12*D*). This suggests that older soils host a richer community of methanogens, enhancing overall demethylation capacity. In summary, we identified six core DMA-demethylating archaea that appear to be the key drivers of DMA breakdown in paddy soils, and their presence strongly predicts a soil's ability to demethylate (and thereby detoxify) DMA.

Metagenome-Assembled Genomes (MAGs) Harboring *arsM* and *mcrA* Genes Linked to Arsenic Methylation and DMA Demethylation. To gain a more holistic view of these organisms, we reconstructed metagenome-assembled genomes (MAGs) from our sequencing data. We assembled 242 high-quality MAGs (genomes ≥50% complete, ≤10% contamination), including 197 bacterial and 17 archaeal genomes (Dataset S1). Among these, we found 69 bacterial MAGs that contained at least one *arsM* gene (potential arsenic methylators) and 13 archaeal MAGs that contained an *mcrA* gene (potential demethylators) (*SI Appendix*, Fig. S13).

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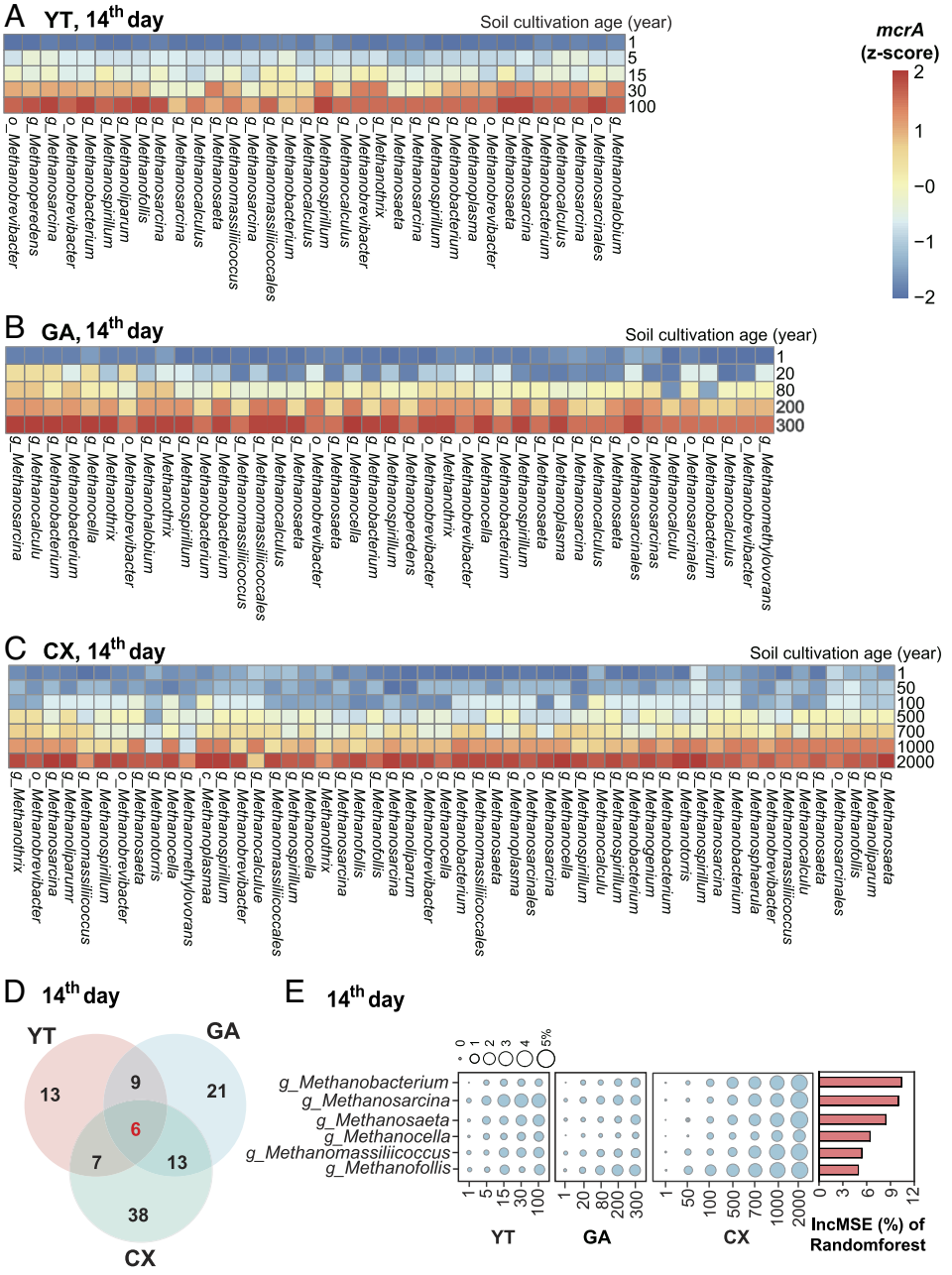


Fig. 5. Core microbial DMA demethylators in three paddy soil chronosequences and their correlations with DMA demethylation rates at the 14th day of flooding. (*A*–*C*) Identified DMA demethylators containing methyl coenzyme M reductase submit A (*mcrA*) genes with abundance positively correlated ($P < 0.05$) with DMA demethylation rates in the YT (*A*), GA (*B*), and CX (*C*) paddy soil chronosequences at the 14th day of flooding. (*D*) Venn diagram showing the number of identified DMA demethylators and the core microbial DMA demethylators common to all three paddy chronosequences. (*E*) Relative abundances of these six core DMA demethylators and their importance to DMA demethylation rates, as determined by a random forest model. (*F* and *G*) Correlations between the total identified DMA demethylators (*F*) and the six core DMA demethylators (*G*) with DMA methylation rates at the 14th day of flooding.

The *arsM*-bearing MAGs spanned multiple phyla (*Acidobacteriota*, *Bacillota*, *Planctomycetota*, *Verrucomicrobiota*, *Pseudomonadota*, etc.), reflecting the diversity noted earlier. The *mcrA*-bearing MAGs all fell within *Euryarchaeota* (methanogens) (Fig. 6 and Dataset S1).

To assess the functional relevance of these MAGs, we evaluated their abundances in relation to *arsM*-driven arsenic methylation rates and *mcrA*-driven DMA demethylation rates. Fifteen *arsM*-harboring MAGs exhibited significant positive correlations with arsenic methylation rates (*SI Appendix*, Fig. S13) and were identified as part of the aforementioned 11 core arsenic methylators (Fig. 4 *D* and *E*). These MAGs were consistently detected in soils of varying cultivation ages across all three paddy soil chronosequences. Their relative abundances increased with soil cultivation age, and taxonomic analysis revealed their distribution across multiple phyla, including *Acidobacteriota*, *Bacillota*, *Bacteroidota*, *Desulfobacterota*,

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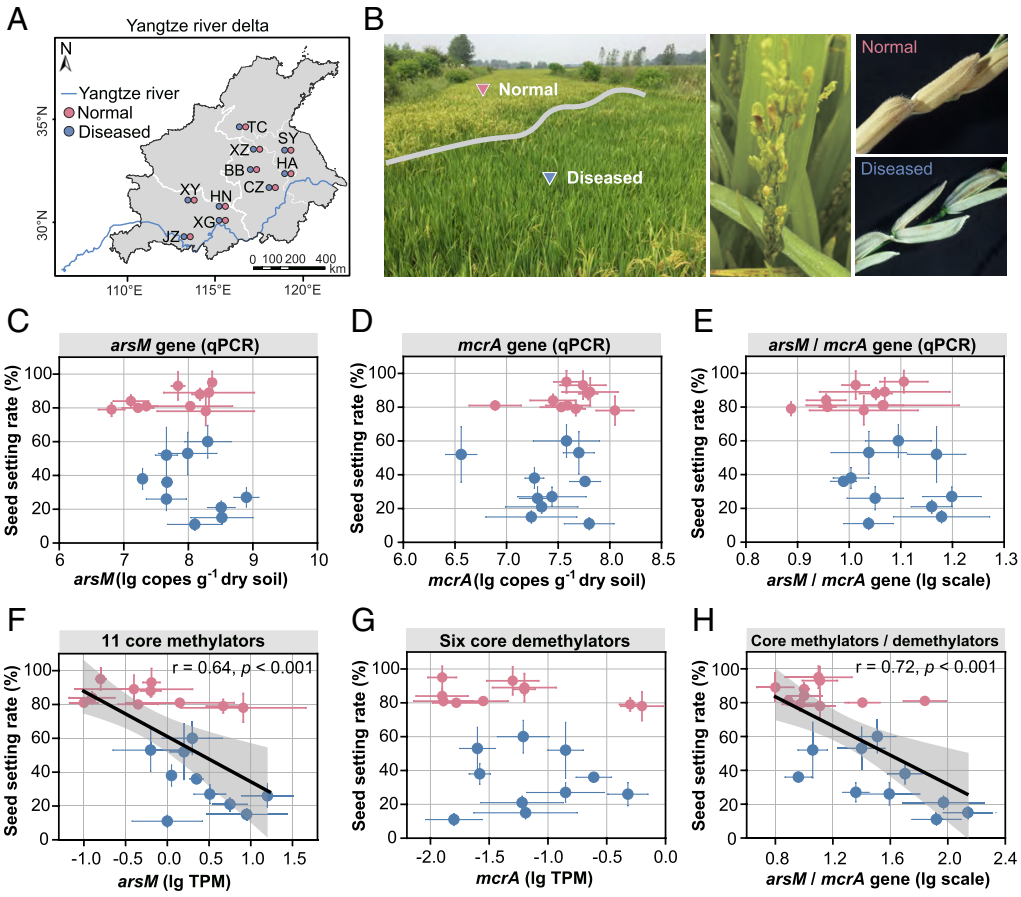


Fig. 6. Field surveys of rice straighthead disease in the middle and lower Yangtze River region, and correlations between disease incidence and the abundances of core methylators or demethylators. (*A*) Map of surveyed sites comparing rice straighthead disease-affected fields with adjacent healthy fields. (*B*) Typical symptoms of rice straighthead disease. (*C–E*) Correlations between seed setting rate and the abundances of the *arsM* gene (*C*), the *mcrA* gene (*D*), and the ratio of *arsM* to *mcrA* gene abundance, as quantified by q-PCR (*E*). (*F–G*) Correlations between seed setting rate and the abundances of the 11 core As methylators (*F*), the six core DMA demethylators (*G*) identified in the three paddy soil chronosequences, and the ratio of core methylators to demethylators (*H*).

Gemmatimonadota, *Methylomirabilota*, *Myxococcota*, *Planctomycetota*, *Pseudomonadota*, and *Verrucomicrobiota*. (*SI Appendix*, Fig. S14*A*).

Similarly, nine methanogenic MAGs harboring *mcrA* genes showed significant positive correlations with DMA demethylation rates (*SI Appendix*, Figs. 13 and S14*B*) and were identified as part of the six core DMA demethylators (Fig. 5 *D* and *E*), likely representing key DMA demethylating archaea. These MAGs were identified among all three chronosequences, with their relative abundances also increasing with soil cultivation age. Taxonomic assignments revealed their affiliations to methanogenic genera, including *Methanobacterium, Methanothrix harundinacea*, Methanobacterium aggregans, *Methanothermobacter, Methanomassiliicoccus, Methanobrevibacter arboriphilus, Methanothrix, Methanocella paludicola SANAE,* and *Methanocella* (*SI Appendix*, Fig. S14*B*).

Microbial Predictors of Straighthead Disease in the Field. Excessive accumulation of DMA in rice is known to cause straighthead disease (15, 16). We therefore investigated whether the core microbial methylators and demethylators we identified are directly linked to straighthead disease incidence. We collected ten paired soil and rice samples from within a single large paddy field in the Yangtze River Delta region, sampling adjacent healthy vs. diseased patches of rice (i.e., plants with normal grain fill vs. plants showing straighthead symptoms) under the same soil conditions (Fig. 6*A*). We performed metagenomic sequencing on these soils and measured straighthead disease severity in the plants (noting distorted husks, "parrot beaks," erect panicles) along with DMA levels in rice husks. The seed setting rate (percentage of grains filled per panicle) was used as a quantitative indicator of disease severity (healthy plants have high seed set, diseased plants have many unfilled grains; Fig. 6*B*).

Using these field samples, we checked multiple microbial indicators against disease outcome. Interestingly, traditional bulk measures like total *arsM* gene or total *mcrA* gene abundance, and even the ratio of total *arsM* to *mcrA* genes (by qPCR), showed no clear correlation with seed setting rate (Fig. 6 *C–E*), suggesting that merely counting all methylation or demethylation genes did not predict disease. However, when we focused on the specific core taxa, a pattern emerged. The combined abundance of the 11 core arsenic methylators had a strong negative correlation with seed setting rate ($r = -0.64$, $P < 0.001$; Fig. 6*F*). This means higher presence of the core DMA-producing microbes was associated with more severe straighthead disease (lower seed set). In contrast, the abundance of the six core demethylators alone showed no significant relationship with seed set (Fig. 6*G*). Notably though, the ratio of core methylators to core demethylators in the soil microbiome gave an even stronger correlation with disease severity ($r = -0.72$, $P < 0.001$; Fig. 6*H*). Soil patches where methylators far outnumbered demethylators had the worst straighthead outcomes. Consistently, we found that rice husk samples from those patches had higher proportions of DMA (*SI Appendix*, Fig. S15), which aligns with severe straighthead. These results suggest that it is specifically the balance of the key microbes—more than just total arsenic or total microbes—that drives straighthead disease. Core arsenic-methylating bacteria appear to promote straighthead disease when not kept in check by demethylating archaea, whereas a balanced or demethylator-dominated community can modulate DMA levels and help prevent the disease.

Global Distribution of Methylating vs. Demethylating Microbes and Arsenic Risk. Finally, we extended our analysis to the global scale. We analyzed metagenomic data from 801 paddy soil samples worldwide, to map the distribution of the core arsenic methylators and demethylators. From each metagenome, we identified *arsM*

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and *mcrA* sequences using stringent criteria (≥60% amino acid identity, ≥70% query coverage, e-value ≤10^{-5}) to capture homologs of the core genes identified above. We then quantified the relative abundance of the 11 core methylators and six core demethylators in each sample (expressed in log-transformed TPM, transcripts per million) and plotted these on world maps (Fig. 7 *A* and *B*). We also calculated their ratio for each sample and mapped that as an indicator of arsenic risk (Fig. 7*C* and Dataset S2).

The global maps reveal substantial geographic variation in these microbial communities (Fig. 7). The relative abundance of core arsenic methylators ranged over ~3 orders of magnitude across regions (Fig. 7*A*), and similarly for core demethylators (Fig. 7*B*). Critically, the methylator-to-demethylator ratio varies widely worldwide (from 0.54 to 2.4; Fig. 7*C*). Certain regions emerge as hotspots with high ratios (i.e., potential high arsenic-methylation pressure). Notable hotspot regions include the Western and Central United States, Southern Europe, Northeastern Japan, Northeastern China, and the Yangtze River Delta (China). These areas show methylator-to-demethylator ratios generally above 1.5 (Fig. 7*C*). In fact, the highest average ratios were observed in the western United States (mean ~1.68) and central United States (~1.67), followed by Southern Europe (~1.59), parts of Japan (~1.39), and parts of Asia (~1.24) (Fig. 7*D*). Within China, Northeast China (mean ~1.59) and the Yangtze River Delta (~1.57) had higher ratios than southern China (~1.20) (Fig. 7 *C* and *D*). These patterns line up with historical paddy ages: Many hotspot regions (Americas, Europe, Northeast Asia) are dominated by relatively recently established rice systems, whereas locations within South Asia or parts of China with millennia-old paddies show lower ratios (consistent with more demethylators).

To explore the relationship between rice grain arsenic speciation and soil methylator-to-demethylator ratios, we first conducted a proximity-based analysis to spatially match soil metagenomic data with rice grain arsenic speciation data from 103 locations in our previous targeted survey of major rice-producing regions in China (*SI Appendix*, Fig. S16*A*) (6). Coordinates for both datasets were aligned under the WGS1984 system and spatial associations were established using the Generate Near Table tool in ArcGIS Pro. This survey revealed that the proportion of DMA in rice grains was significantly higher in the Northeast and Yangtze River Delta compared to Southern China (*SI Appendix*, Fig. S16*B*). Similarly, the core methylator-to-demethylator ratio was higher in these regions compared to Southern China (*SI Appendix*, Fig. S16*C*). Notably, the proportion of DMA in total rice arsenic concentrations exhibited a highly significant positive correlation with the core methylator-to-demethylator ratio ($r = 0.76$, $P < 0.001$), which was much stronger than the correlation observed with the ratio of *arsM* to *mcrA* genes measured via qPCR ($r = 0.13$, $P = 0.053$) (*SI Appendix*, Fig. S16 *D* and *E*).

We then expanded to a global analysis: We compiled worldwide rice grain arsenic speciation data from 1,103 samples spanning 462 locations on different continents (from literature sources). Using a similar proximity-matching approach, we linked each rice grain data point to nearby soil microbiome samples in our 801-sample dataset (Fig. 7*E* and Dataset S2). Globally, we found a strong positive correlation between grain arsenic species and the core methylator-to-demethylator ratios in soil ($r = 0.78$, $P < 0.001$; Fig. 7*E*). These results indicate that the global pattern of rice arsenic speciation is largely driven by the methylator-to-demethylator balance in paddy soils.

Discussion

Our study reveals that the microbial ecology of paddy soils, shaped by cultivation history, governs arsenic speciation in rice and the risk of straighthead disease. Specifically, paddy cultivation age

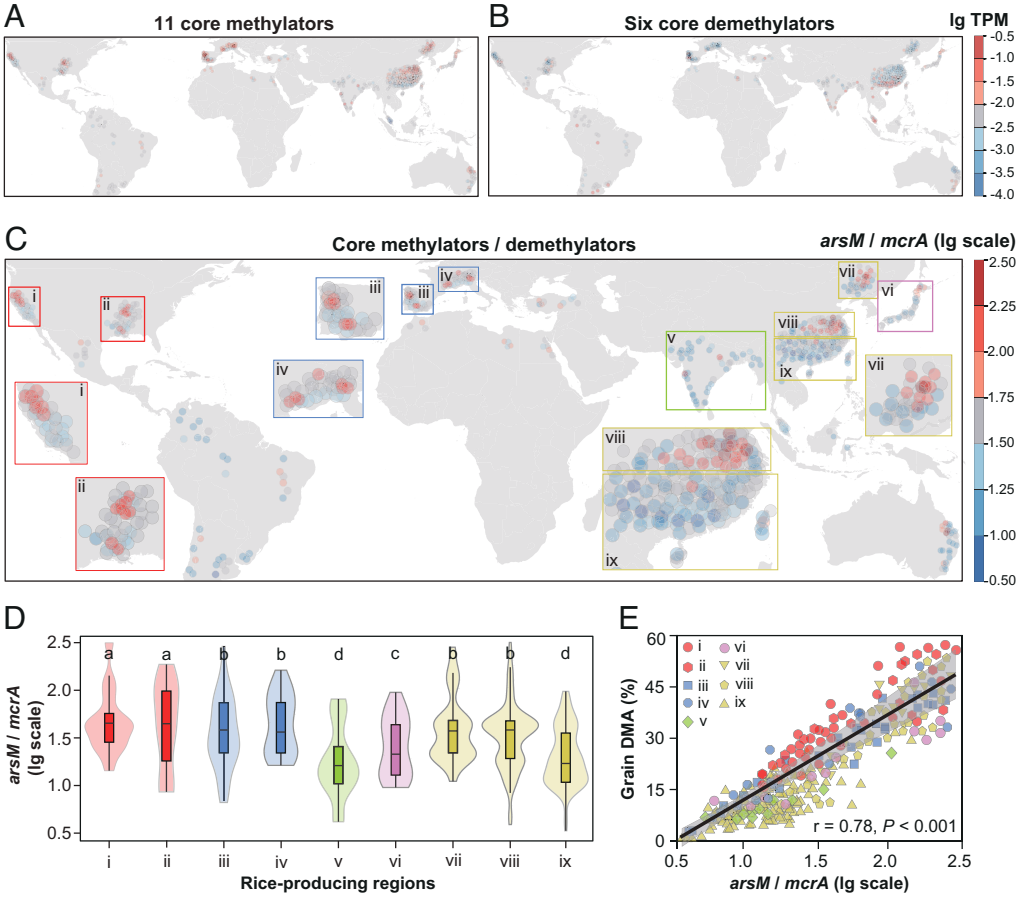


Fig. 7. Global distribution of core arsenic methylators and DMA demethylators in paddy soils, and their association with DMA levels in rice grain. (*A*–*C*) Geographic distributions of the 11 core methylators' abundance in paddy soils worldwide (*A*), the six core demethylators' abundance (*B*), and the ratio of core methylators to demethylators (*C*) in global paddy soils. (*D*) Regional variations in the ratio of core methylators to demethylators across different rice-producing regions. (*E*) Correlation between the proportion of DMA in rice grains and the ratio of core methylators to demethylators globally.

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modulates the balance between arsenic-methylating and demethylating microorganisms, determining the accumulation of DMA (and its toxic derivatives) in rice grains. By identifying the microbial drivers of arsenic methylation vs. demethylation and their evolution over time, we provide a mechanistic framework for understanding global patterns in rice arsenic speciation and straighthead disease prevalence.

In young and moderate-age paddy soils (<700 y), arsenic-methylating microbes dominated (Fig. 4), driven by cultivation-induced increases in DOC (39, 40), shifts in redox potential (31), and changes in Fe availability (41) (*SI Appendix*, Fig. S5*A* and Fig. 3 *E* and *F*). SOC accumulation with cultivation age, facilitated by plant residues and Fe oxides (42, 43), fueled microbial activity and lowered redox potential, creating the anoxic conditions that favored arsenic methylation. Concurrently, microbial Fe(III) and As(V) reduction enhanced the mobilization of As(III) (42), the key substrate for microbial arsenic methylation. The positive correlation between porewater Fe(II) and As(III) further supports overlapping roles of Fe- and As-reducing microbes. Notably, *arsM* genes were predominantly distributed among anaerobic microbes (e.g., 75% of MAGs carrying *arsM* genes lacked cytochrome C oxidase genes), including sulfate-reducing bacteria (SRB), which are more abundant in paddy soils than in upland soils (44) and are responsive to changes in paddy soil cultivation age. Among the 11 core As methylators, three *arsM* hosts (*Desulforamulus*, *Desulfallas*, and *Desulfitobacterium*), all classified as SRB, exhibited a significant increase in abundance during the first 700 y of cultivation, contributing to enhanced of arsenic methylation and DMA accumulation. However, beyond 700 y, stabilization of methylating microbes tempered further increases in DMA levels, indicating a more balanced microbial system.

Conversely, older paddy soils exhibited a progressive enrichment of methanogenic archaea carrying the *mcrA* gene, which enhanced DMA demethylation capability (Fig. 5). Methylotrophic methanogens, key contributors to DMA demethylation (24, 32), were more abundant in paddy soils compared to upland soils (45, 46) and demonstrated high sensitivity to cultivation age (45). Among the six core demethylators identified, two genera (*Methanomassiliicoccus* and *Methanosarcina*) that are both methylotrophic methanogens increased consistently in abundance with cultivation age, highlighting their critical role in enhancing DMA demethylation during long-term soil development. Cultivation of paddy rice involves flooding-draining cycles, and paddy soils are usually not flooded outside the rice growing period. This may explain the slow evolution of microbial communities, especially the slow buildup of methanogens, which are strict anaerobes. The expansion of methanogenic archaea in older soils is likely driven by the gradual evolution of redox conditions and carbon substrate availability. Specifically, with prolonged flooding, older soils exhibit more rapid and sustained decreases in redox potential (Eh), accompanied by more complete depletion of alternative electron acceptors such as sulfate (*SI Appendix*, Fig. S2). In parallel, the accumulation of labile organic carbon supported syntrophic microbial interactions and fuels methanogenesis, enabling continuous enrichment of demethylating archaea (35, 47). As a result, the rate of DMA demethylation increased with cultivation age. Collectively, these shifts caused net DMA accumulation—the balance between methylation and demethylation—to peak at around 700 y, after which it declined, reflecting a dynamic and evolving microbial balance shaped by the legacy of soil cultivation.

Our findings are consistent with biodiversity-ecosystem functioning (BEF) theory (48–50), which posits that higher biodiversity enhances functional outcomes (50). While *arsM* gene diversity showed no clear relationship with cultivation age or arsenic methylation rates (*SI Appendix*, Fig. S9), *mcrA* gene diversity increased significantly and correlated strongly with demethylation rates (*SI Appendix*, Fig. S12). These results suggest that cultivation age promotes the diversity of arsenic demethylators, leading to enhanced DMA demethylation and reduced accumulation over time.

The identification of 11 core arsenic methylators and six DMA demethylators underscores their evolutionary and ecological importance in arsenic cycling in paddy soils (Figs. 4 and 5). These microbial groups exhibit significant shifts in abundance with soil cultivation age and are broadly distributed across global paddy soils. Notably, the strong correlation between the core methylator-to-demethylator ratio and DMA levels in global rice grains (Fig. 7) establishes this ratio as a robust predictive tool for assessing DMA accumulation risks on a global scale. Furthermore, field surveys revealed a direct link between elevated methylator-to-demethylator ratios and both increased incidence of straighthead disease in paddies (Fig. 6*H*) and higher proportions of DMA in rice husks (*SI Appendix*, Fig. S15). These findings emphasize the importance of microbial balance in preventing disease outbreaks, particularly in newly converted paddies that inherently exhibit higher methylator-to-demethylator ratios.

These findings also provide a mechanistic explanation for the geographic variation in rice arsenic speciation. While DMA poses less toxicity to humans, its derivative, DMMTA, is highly toxic and shows a strong positive correlation to DMA concentration (6). Regions with shorter rice cultivation histories (e.g., Europe and the Americas) had higher methylator-to-demethylator ratios, resulting in elevated DMA and DMMTA levels in rice grains (Fig. 7 *C* and *D*). In contrast, ancient Southeast Asian paddies with rich demethylator communities showed reduced accumulation and disease risk.

Practically, the methylator-to-demethylator ratio serves as a reliable biomarker for toxic arsenic accumulation and straighthead disease risk. A threshold ratio of >1.5 indicates elevated straighthead disease risk, providing a basis for targeted management decisions, such as midseason draining and use of silicon fertilizers (51, 52), which are effective to alleviate straighthead disease.

In conclusion, this work shows that paddy cultivation age fundamentally alters soil microbial communities in ways that directly influence crop health and food safety. By linking microbial ecology, arsenic biogeochemistry, and plant pathology, we explain long-observed geographic patterns in rice arsenic species and straighthead disease risk. Our identification of specific microbial taxa as predictive indicators, combined with a global risk map, offers actional tools for policy and practice. These insights can inform preemptive strategies in emerging rice-growing regions—especially across Africa and South America—to avoid arsenic-related pitfalls and promote sustainable, resilient food systems.

Materials and Methods

Study Sites and Sample Collection. Three paddy soil chronosequences were selected from China: Cixi (CX), Guang'an (GA), and Yingtan (YT) (Fig. 1*A*). The CX chronosequence originated from coastal salt marshes reclaimed over 2,000 y through dike constructions (53), with cultivation durations of 1, 50, 100, 500, 700, 1,000, and 2,000 y (53). The GA chronosequence comprises terraced paddy fields with soil cultivation ages increasing from the summit (1 y) to the base of the hill (300 y), comprising five ages: 1, 20, 80, 200, and 300 y (54). The YT chronosequence is a river delta region where paddy soils were first reclaimed from forested land near a river, with cultivation ages ranging from 1 to 100 y (1, 5, 15, 30, and 100 y) (55). At rice maturity, 16 rice panicle and 17 soil samples were collected. Soils were air-dried for physicochemical analysis (*SI Appendix*, Table S1) and used in incubation experiments. Soil properties were measured following standard methods (56).

Anaerobic Microcosm Incubation. To assess arsenic transformation over time, anaerobic microcosms were established. Air-dried soils (30 g) were placed into 100 mL glass serum bottles with 30 mL deionized water, purged

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with N_2, sealed, and incubated at 25 ± 1 °C for 28 d. Triplicate bottles were sacrificed at 1, 7, 14, 21, and 28 d. CH_4 in the headspace was measured by gas chromatography. Redox potential (Eh) was recorded in a N_2-filled glove box. Slurries were shaken and settled for 30 min, then supernatants were filtered (0.22 μm) and divided into three parts: The first was analyzed immediately for sulfide (57); the second preserved with 10 mM DTPA for total arsenic and species analysis by HPLC-ICP-MS (Flexar and NexION 300X, PerkinElmer); the third acidified with HCl for DOC and total dissolved Fe analysis. Soils were stored at −80 °C for DNA extraction.

Total Arsenic and Arsenic Species Analysis. For total arsenic, rice grains (0.25 g) were digested with HNO_3, and husks (0.1 g) with HNO_3:$HClO_3$ (85:15 v/v), then analyzed via ICP-MS. Arsenic species in husks were extracted with 1% HNO_3 and in grains via enzymatic digestion (6, 31). DMA and methylated thioarsenates in grain and porewater were separated on a C18 column (Atlantis dC18, Waters) with 20 mM $NH_4H_2PO_4$ (pH 3.0) (6). Oxyarsenates (iAs and DMA) were separated using an PRP-X100 anion-exchange column with 8.5 mM NH_4NO_3/$NH_4H_2PO_4$ buffer (pH 6.0) as the eluent (31).

Arsenic Methylation and DMA Demethylation Rates. Arsenic methylation and DMA demethylation rates were quantified using kinetic models (as detailed in *SI Appendix*, Fig. S4), based on Michaelis–Menten kinetics:

$$V_1 = V_{max1} \cdot [C_{iAs}]/(K_{m1} + [C_{iAs}]), \quad [1]$$

$$V_2 = V_{max2} \cdot [C_{DMA}]/(K_{m2} + [C_{DMA}]). \quad [2]$$

Here, V_1 and V_2 (μg L^{-1} d^{-1}) represent methylation and demethylation rates, respectively. V_{max1} and V_{max2} (μg L^{-1} d^{-1}) are the maximum methylation and demethylation rates. C_{iAs} (μg L^{-1}) and C_{DMA} (μg L^{-1}) denote substrate concentrations of iAs and DMA in soil porewater. K_{m1} and K_{m2} (μg L^{-1}) are the Michaelis constants for methylation and demethylation, respectively.

The maximum rates V_{max1} and V_{max2} were defined as

$$V_{max1} = k_1 \cdot [arsM], \quad [3]$$

$$V_{max2} = k_2 \cdot [mcrA]. \quad [4]$$

Here, k_1 and k_2 [μg L^{-1} d^{-1} (copes g^{-1} soil)$^{-1}$] are catalytic constants, and [*arsM*] (copes g^{-1} soil) and [*mcrA*] (copes g^{-1} soil) are the concentrations of functional genes driving methylation and demethylation, respectively.

The net DMA accumulation rate V_3 (μg L^{-1} d^{-1}) was calculated as

$$V_3 = V_1 - V_2. \quad [5]$$

The DMA concentration C_{DMA} (μg L^{-1}) in porewater over time was expressed as

$$C_{DMA} = \int_0^n V_3 dt = \int_0^n [k_1[arsM][C_{iAS}]/(K_{m1} + [C_{iAS}]) - k_2[mcrA][C_{DMA}]/(K_{m2} + [C_{DMA}])]dt. \quad [6]$$

Parameters k_1 and k_2, along with methylation, demethylation, and net accumulation rates at the 7th and 14th days of incubation, were calculated using MATLAB software. Observed accumulation rates (calculated as the derivative of the DMA accumulation curves) at the 7th and 14th days (*SI Appendix*, Fig. S5 *A* and *B–D*) were compared to rates calculated from Eq. **5** to assess the model's accuracy.

DNA Extraction and Quantitative Real-Time PCR. Total soil genomic DNA was extracted using the DNeasy Powersoil kit (QIAGEN, Germany). The concentration and purity of the extracted DNA were assessed with a NanoDrop 2000C spectrophotometer (Thermo Scientific, Wilmington). Quantifications of target genes, including bacterial 16S rRNA, archaeal 16S rRNA, *arsM, mcrA*, *dsr*, and *arrA* genes, were performed using a Real-Time PCR detection System (Bio-Rad CFX96). Primer details are provided in *SI Appendix*, Table S6.

Metagenomic Sequencing and Analysis. Total soil DNA were extracted and sequenced on the Illumina NovaSeq platform (paired-end, 400 bp inserts) using the NEXTFLEX Rapid DNA-Seq kit. Quality control was performed with fastp (58) (v0.21.0), yielding ~8.07 billion reads (average ~79 million/sample). Clean reads were assembled with MEGAHIT (59) (v1.2.9), and contigs of ≥300 bp were used for gene prediction with Prodigal (60) (v2.6.3). Nonredundant open reading frames (ORFs) were clustered at 95% identity using CD-HIT (61) (v4.8.1), and gene abundances were quantified in transcripts per million (TPM) using Salmon (62) (v1.10.2).

Contigs were binned into metagenome-assembled genomes (MAGs) using metaWRAP (63) (v1.3.2), integrating MaxBin2 (64), MetaBAT (65), and MetaBAT2. Bins were refined, deduplicated using dRep (66) (v3.4.3), and quality-checked with CheckM (67) (v1.2.2); only MAGs with ≥ 50% completeness and ≤ 10% contamination were retained. MAG abundance was calculated by read coverage, and taxonomy was assigned via GTDB-Tk (68) (v2.1.1). Phylogenetic trees were generated with PhyloPhlAn (v3.1.68) and visualized using iTOL. ORFs within MAGs were predicted using Prodigal.

Diversity and Abundance of *arsM* and *mcrA* Genes. To assess the diversity and abundance of the *arsM* and *mcrA* genes, we analyzed the predicted ORFs from both contigs and MAGs. These ORFs were compared against two specialized databases: an *ArsM* database (31) and the McycDB database (69) using BLASTp (70). We applied stringent criteria for sequence alignment, retaining hits with an amino acid identify of ≥60%, a query sequence length coverage of ≥ 70%, and an e-value ≤ 10^{-5}. This approach ensured the accurate identification and quantification of *arsM* and *mcrA* genes within the metagenomic datasets.

Surveys of Rice Straighthead Disease-Affected Fields and Adjacent Healthy Fields. Field surveys were conducted in ten sites across Anhui, Jiangsu, Shandong, and Hubei provinces in China, each comprising a straighthead-affected and an adjacent healthy rice field. From each field, three replicates of 6 to 20 panicles and corresponding soil samples were collected for metagenomic and arsenic speciation analyses. To assess disease severity, the seed setting rate was calculated as the percentage of filled grains among all grains, with filled and unfilled grains separated for this analysis. Grain samples were dried at 60 °C for 2 d and husk arsenic species were quantified via HPLC-ICP-MS.

Metagenomic Analysis of Global Paddy Soils. A total of 1,021 soil metagenomes were retrieved from the European Nucleotide Archive (ENA; https://ebi.ac.uk/ena/) and the Sequence Read Archive (https://ncbi.nlm.nih.gov/sra/). To minimize bias, we selected only uncultured, untreated control soil samples sequenced using Illumina paired-end shotgun platforms (FASTQ format) with an average read length >100 bp, excluding single-end reads and non-Illumina technologies. After filtering, 801 metagenomes from 462 global paddy sites (including 27 from this study) were retained. All reads were assembled into contigs, and ORFs were predicted for downstream analysis.

Statistical analysis. Statistical analyses were performed using R and SPSS. Differential taxa were visualized with the pheatmap package, and environmental drivers were evaluated via random forest models. Group differences were assessed by one-way ANOVA with Tukey's test ($P < 0.05$). Correlation, regression, and PCA were conducted in IBM SPSS, and spatial maps were generated using ArcGIS 10.2.

Data, Materials, and Software Availability. Metagenomic raw sequencing data have been deposited in National Center for Biotechnology Information (NCBI) Sequence Read Archive (SRA) (PRJNA1068274) (71).

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